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Training Multi-Bit Quantized and Binarized Networks with a Learnable Symmetric Quantizer

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ABSTRACT Quantizing weights and activations of deep neural networks is essential for deploying them in resource-constrained devices, or cloud platforms for at-scale services. While binarization is a special case of quantization, this extreme case often leads to several training difficulties, and necessitates specialized models and training methods. As a result, recent quantization methods do not provide binarization, thus losing the most resource-efficient option, and quantized and binarized networks have been distinct research areas. We examine binarization difficulties in a quantization framework and find that all we need to enable the binary training are a symmetric quantizer, good initialization, and careful hyperparameter selection. These techniques also lead to substantial improvements in multi-bit quantization. We demonstrate our unified quantization framework, denoted as UniQ, on the ImageNet dataset with various architectures such as ResNet-18, -34 and MobileNetV2. For multi-bit quantization, UniQ outperforms existing methods to achieve the state-of-the-art accuracy. In binarization, the achieved accuracy is comparable to existing state-of-the-art methods even without modifying the original architectures.

INDEX TERMS Learnable quantizer, quantization, binarization, model compression, machine learning, neuromorphic computing.

I. INTRODUCTION

Deep neural networks have achieved tremendous success in various fields including computer vision [31], natural language processing [52], and speech recognition [8], having demonstrated unprecedented predictive performance. However, the computational complexity and memory access count required by the existing models pose a challenge in deploying them to resource-constrained devices, and applying to latency-critical services. To address this challenge, efficient network architectures, manually designed [46] or automatically searched [48], and model compression techniques such as pruning [17] and quantization [5], [24], [47], [56] have been studied. In practical model deployment, quantization is a necessary step, and often the last means to control the performance and efficiency trade-off once the model architecture is fixed [24].

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Quantizing weights and activations to a lower precision not only reduces the computational complexity but also the model size, memory footprint, and memory access count but at cost of degraded performance. Recent quantization methods [11], [29] can overcome accuracy degradation from their full-precision counterparts even with 4-bit weights and activations. However, most of these methods do not present 1-bit results, possibly due to their severely degraded performance or because 1-bit training does not converge, ditching the most efficient option.

While Genc *et al.* [57] considered binarized neural networks together with multi-bit quantization, binarized neural networks have been a distinct research topic from the quantized models. Most studies on binarized models [7], [23], [37], [44] focused on the 1-bit case solely. Binarized networks have gained attention because of the expensive floating-point multiplications and additions being replaced by efficient xnor and popcount operations. However, this is not specific to binarized networks, and any low-precision networks can be executed efficiently by such operations as shown